UNIVERSITY NAME

COURSE NAME — Final Exam

Year: 2025

Exercise 1:

Let $E = C^1([0,1],\mathbb{R})$ be the space of continuously differentiable functions and $\|\cdot\|_\infty$ the uniform convergence norm. Define

$$N(f) = |f(0)| + \|f'\|_\infty \quad \text{and} \quad N'(f) = \|f\|_\infty + \|f'\|_\infty.$$

1. Show that N and N' are norms on E.
2. Show that N and N' are equivalent.
3. Are the norms N and N' equivalent to the norm $\|\cdot\|_\infty$?

Answer Area

Let $E = C^1([0,1],\mathbb{R})$, $\|\cdot\|_\infty$ the sup norm.

1. Show N and N' are norms

For $N(f) = |f(0)| + \|f'\|_\infty$:

- $N(f) = 0 \Rightarrow |f(0)| = 0$ and $\|f'\|_\infty = 0 \Rightarrow f(0) = 0$ and $f' = 0 \Rightarrow f = 0$.
- $N(\lambda f) = |\lambda f(0)| + \|\lambda f'\|_\infty = |\lambda|(|f(0)| + \|f'\|_\infty) = |\lambda| N(f)$.
- $N(f+g) = |f(0)+g(0)| + \|f'+g'\|_\infty \leq |f(0)| + |g(0)| + \|f'\|_\infty + \|g'\|_\infty = N(f) + N(g)$.

So N is a norm.
For $N'(f) = \|f\|_\infty + \|f'\|_\infty$:

- $N'(f) = 0 \Rightarrow \|f\|_\infty = 0$ and $\|f'\|_\infty = 0 \Rightarrow f = 0$.
- $N'(\lambda f) = \|\lambda f\|_\infty + \|\lambda f'\|_\infty = |\lambda|(\|f\|_\infty + \|f'\|_\infty) = |\lambda| N'(f)$.
- $N'(f+g) \leq \|f+g\|_\infty + \|f'+g'\|_\infty \leq \|f\|_\infty + \|g\|_\infty + \|f'\|_\infty + \|g'\|_\infty = N'(f) + N'(g)$.

So N' is a norm.

2. Show N and N' are equivalent

For $f \in E$, $x \in [0,1]$:

$$|f(x)| = \left| f(0) + \int_0^x f'(t)dt \right| \leq |f(0)| + \|f'\|_\infty = N(f).$$

Taking supremum: $\|f\|_\infty \leq N(f)$. Also $|f(0)| \leq \|f\|_\infty$.
Thus:

$$N(f) = |f(0)| + \|f'\|_\infty \leq \|f\|_\infty + \|f'\|_\infty = N'(f) \leq N(f) + \|f'\|_\infty.$$

Actually better: $N(f) \leq N'(f)$ and $N'(f) = \|f\|_\infty + \|f'\|_\infty \leq N(f) + N(f) = 2N(f)$.
So $N(f) \leq N'(f) \leq 2N(f)$. Hence N and N' are equivalent.

3. Are N, N' equivalent to $\|\cdot\|_\infty$?

No. Consider $f_n(x) = x^n$ on $[0,1]$.
$\|f_n\|_\infty = 1$ for all n, but $N(f_n) = |f_n(0)| + \|f_n'\|_\infty = 0 + n = n \to \infty$.
So $\frac{N(f_n)}{\|f_n\|_\infty} = n \to \infty$, thus N and $\|\cdot\|_\infty$ are not equivalent.
Since N and N' are equivalent, N' is also not equivalent to $\|\cdot\|_\infty$.

Exercise 2:

Let $(E, \|\cdot\|)$ be a normed vector space.

1. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence converging to x in E and let K be a non-empty subset of E defined by:
$$K = \{x_n \mid n \in \mathbb{N}\} \cup \{x\}.$$
   (a) Show that K is compact in E using the definition of compactness.
   (b) Show that K is compact in E using the Borel property.
2. Let $(A_n)_{n\in\mathbb{N}}$ be a decreasing sequence of non-empty compact subsets of E. Show that the intersection $A = \bigcap_{n\in\mathbb{N}} A_n$ is a non-empty compact subset of E.

Answer Area

1. $K = \{x_n : n \in \mathbb{N}\} \cup \{x\}$, $x_n \to x$
   (a) Using definition
   We show every sequence in K has a convergent subsequence in K.
   Let (y_k) be a sequence in K. If $y_k = x$ infinitely often, we have constant subsequence. Otherwise, (y_k) contains infinitely many x_n. Since $x_n \to x$, we can extract subsequence converging to $x \in K$.
   (b) Using Borel property
   Let $\{U_i\}_{i\in I}$ be an open cover of K. Since $x \in K$, $\exists i_0$ with $x \in U_{i_0}$. As $x_n \to x$, $\exists N$ such that $\forall n \geq N$, $x_n \in U_{i_0}$. The remaining finite set

$\{x_1, \ldots, x_{N-1}\}$ can be covered by finitely many U_i. Thus K is covered by finitely many U_i.

2. A_n decreasing non-empty compact sets, $A = \bigcap_{n\in\mathbb{N}} A_n$

   Choose $x_n \in A_n$. Since $A_n \subset A_0$ and A_0 compact, (x_n) has convergent subsequence $x_{\varphi(n)} \to \ell$.

   For fixed p, for n large, $\varphi(n) \geq p$, so $x_{\varphi(n)} \in A_{\varphi(n)} \subset A_p$. Since A_p closed, $\ell \in A_p$. This holds $\forall p$, so $\ell \in \bigcap A_n = A$. Thus $A \neq \emptyset$.

   A is closed (intersection of closed sets) and contained in compact A_0, so A is compact.

Exercise 3:

Define the function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ by:

$$f(x,y) = \begin{cases} \dfrac{x^2y^3}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

1. Compute $\lim_{(x,y)\to(0,0)} f(x,y)$.
2. Show that f is continuous on $\mathbb{R}^2$ (justify your answer).
3. (a) Determine $df_{(x,y)}$, the differential of f for $(x,y) \neq (0,0)$.

   (b) Compute $\frac{\partial f}{\partial x}(0,0)$ and $\frac{\partial f}{\partial y}(0,0)$.
4. Show that f is of class C^1 on $\mathbb{R}^2$.
5. Let g be a function defined on $\mathbb{R}^2$ by:

$$g(x,y) = f(x,y) - \frac{1}{2}.$$

   (a) Show that there exists a neighborhood U of 1 and a C^1 function μ on U such that:

$$\mu(1) = 1 \quad \text{and} \quad \forall x \in U, \quad g(x, \mu(x)) = 0.$$

   (b) Compute $\mu'(1)$, the derivative of μ at the point 1.

Answer Area

1. Limit at $(0,0)$ For $(x,y) \neq (0,0)$:

$$|f(x,y)| = \frac{x^2|y|^3}{x^2+y^2} = x^2|y| \cdot \frac{y^2}{x^2+y^2} \leq x^2|y| \cdot 1 \leq (x^2+y^2)^{3/2}.$$

   As $(x,y) \to (0,0)$, $|f(x,y)| \to 0$. So $\lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$.

2. Continuity on $\mathbb{R}^2$

f is rational on $\mathbb{R}^2 \setminus \{(0,0)\}$, so continuous there. From part 1, continuous at $(0,0)$. Thus continuous everywhere.

3. Differential

   (a) For $(x,y) \neq (0,0)$:

   $$\frac{\partial f}{\partial x} = \frac{2xy^3(x^2+y^2) - x^2y^3 \cdot 2x}{(x^2+y^2)^2} = \frac{2xy^5}{(x^2+y^2)^2},$$

   $$\frac{\partial f}{\partial y} = \frac{3x^2y^2(x^2+y^2) - x^2y^3 \cdot 2y}{(x^2+y^2)^2} = \frac{3x^4y^2 + x^2y^4}{(x^2+y^2)^2}.$$

   Thus $df_{(x,y)}(h,k) = \frac{\partial f}{\partial x}h + \frac{\partial f}{\partial y}k$.

   (b) At $(0,0)$:

   $$\frac{\partial f}{\partial x}(0,0) = \lim_{t\to 0} \frac{f(t,0) - f(0,0)}{t} = \lim_{t\to 0} \frac{0}{t} = 0.$$

   $$\frac{\partial f}{\partial y}(0,0) = \lim_{t\to 0} \frac{f(0,t) - f(0,0)}{t} = \lim_{t\to 0} \frac{0}{t} = 0.$$

4. C^1 on $\mathbb{R}^2$

   Need to show partial derivatives are continuous at $(0,0)$.

   For $\frac{\partial f}{\partial x}$:

   $$\left|\frac{2xy^5}{(x^2+y^2)^2}\right| \leq 2|x||y| \cdot \frac{y^4}{(x^2+y^2)^2} \leq 2|x||y| \to 0.$$

   So $\lim_{(x,y)\to(0,0)} \frac{\partial f}{\partial x}(x,y) = 0 = \frac{\partial f}{\partial x}(0,0)$.

   For $\frac{\partial f}{\partial y}$:

   $$\left|\frac{3x^4y^2 + x^2y^4}{(x^2+y^2)^2}\right| \leq \frac{3x^4y^2}{(x^2+y^2)^2} + \frac{x^2y^4}{(x^2+y^2)^2} \leq 3y^2 + x^2 \to 0.$$

   So $\lim_{(x,y)\to(0,0)} \frac{\partial f}{\partial y}(x,y) = 0 = \frac{\partial f}{\partial y}(0,0)$.

   Thus $f \in C^1(\mathbb{R}^2)$.

5. Implicit function

   (a) $g(x,y) = f(x,y) - \frac{1}{2}$

   At $(1,1)$: $f(1,1) = \frac{1^2 \cdot 1^3}{1^2+1^2} = \frac{1}{2}$, so $g(1,1) = 0$.$\frac{\partial g}{\partial y}(1,1) = \frac{\partial f}{\partial y}(1,1) = \frac{3\cdot 1^4 \cdot 1^2 + 1^2 \cdot 1^4}{(1^2+1^2)^2} = \frac{3+1}{4} = 1 \neq 0$ By implicit function theorem, $\exists U \ni 1$, C^1 function $\mu : U \to \mathbb{R}$ with $\mu(1) = 1$ and $g(x, \mu(x)) = 0$ for $x \in U$.

   (b) Compute $\mu'(1)$

   From $g(x,\mu(x)) = 0$, differentiate:

   $$\frac{\partial f}{\partial x}(1,1) + \frac{\partial f}{\partial y}(1,1) \cdot \mu'(1) = 0.$$

   Now $\frac{\partial f}{\partial x}(1,1) = \frac{2\cdot 1 \cdot 1^5}{(1^2+1^2)^2} = \frac{2}{4} = \frac{1}{2}$.

   So $\frac{1}{2} + 1 \cdot \mu'(1) = 0 \Rightarrow \mu'(1) = -\frac{1}{2}$.